Exercise 8.1

17. 0.008 18. 0.9487 19. 0.2739 20. 0.025

Exercise 8.2

14. $|z| = 1.667$ 15. 1.2308 16. $|z| = 5$ 17. 3.536

Exercise 8.3

1	2	3	4	5	6	7	8	9	10
(a)	(b)	(a)	(b)	(b)	(a)	(c)	(b)	(c)	(a)
11	12	13	14	15	16	17	18	19	20
(a)	(a)	(a)	(d)	(b)	(b)	(a)	(c)	(a)	(c)

Miscellaneous problems

5. 0.015 6. (a) (66.86, 68.04) (b) (66.67,68.22) 7. $|z| = 2.67$

9. Applied Statistics

Exercise 9.1

13. Seasonal Indices

	I	II	III	IV
Total	18.6	20.8	18.8	20.8
Average	3.72	4.16	3.76	4.16
Seasonal indices	94.1772	105.3165	95.1899	105.3165

The Grand Average = 3.95

14. Three yearly moving average

Year	1987	1988	1989	1990	1991	1992
Three yearly moving total	46410	52010	63040	79470	94050	102450
Three yearly moving average	15470	17336.666	21013.333	26490	31350	34150

15. Five yearly moving average

Year	1981	1982	1983	1984	1985	1986	1987	1988
Four yearly moving total	619	617	624	621	615	619	613	606
Four yearly moving average	123.8	123.4	124.8	124.2	123	123.8	122.6	121.2

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